# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

# MATHEMATICS

Past Paper 2023 • Paper Code: 10231

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The formula $\Sigma x/n$ determines .....

(A) arithmetic mean
(B) geometric mean
(C) mode
(D) median

Q2. If $\pi/2 < \theta < \pi$ then '$\theta$' belongs to ..... quadrant.

(A) 4th
(B) 3rd
(C) 2nd
(D) 1st

Q3. $\sin 60° =$ .....

(A) $\cos 30°$
(B) $\cos 60°$
(C) $\tan 30°$
(D) $\cos 90°$

Q4. Perpendicular from the centre of a circle on a chord ..... it.

(A) intersects
(B) trisects
(C) bisects
(D) divides

Q5. If two circles touch internally, the distance between their centres is the ..... of their radii.

(A) sum
(B) difference
(C) product
(D) multiple

Q6. The measure of the semi-circle is .....

(A) 180° (B) 360°

(C) 90° (D) 45°

Q7. An angle whose vertex is on the circle and whose sides contain chords of the circle is ..... angle.

(A) escribed (B) right

(C) inscribed (D) circumscribed

Q8. All those values of the variables for which the given equation is true are called .....

(A) roots (B) factors

(C) absolute values (D) square roots

Q9. If 'w' is the complex cube root of unity then $w^{24}$ = .....

(A) w (B) $w^2$

(C) 1 (D) 0

Q10. Sum of the roots of $4x^2-4x-3=0$ is .....

(A) $-3/4$ (B) $3/4$

(C) $-4/3$ (D) 1

Q11. Comparison of two quantities of the same kind and same unit is called .....

(A) proportion (B) variation

(C) rate (D) ratio

Q12. $1/(x^2-1)$ = .....

(A) $A/(x-1) + B/(x+1)$ (B) $(AX+B)/(x^2-1)$

(C) $A/(x-1) + (BX+C)/(x^2-1)$ (D) none of these

Q13. If A, B and C are any three sets then $A\cap(B\cap C) = (A\cap B)\cap C$ is ..... property of intersection.

(A) associative (B) commutative

(C) distributive (D) trichotomy

Q14. If set 'A' has 2 elements and set B has 3 elements then the number of binary relations in $A\times B$ = .....

(A) $2\times3$ (B) $2^3$

(C) $2^6$ (D) $2^2$

Q15. ..... divides the data in two equal parts.

(A) mean (B) median

(C) mode (D) variance

## SECTION-B

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 36

Q1. Attempt any nine of the following short questions. Each question carries 4 marks.

(i) Solve by factorization and check your answer: $2x^2+2x-11=1$

(ii) Solve $(x+1/x)^2 -10(x+1/x) +16=0$

(iii) Show that the sum of cube roots of unity is zero. i.e. $1+\omega+\omega^2=0$

(iv) Find the quadratic equation whose roots are $3+\sqrt{2}$, $3-\sqrt{2}$

(v) If y varies jointly as x and z, and y=12 when x=9 and z=3, find z when y=6 and x=15.

(vi) Prove that each fraction in $(x+y)/(a+b) = (y+z)/(b+c) = (z+x)/(c+a)$ is equal to $(x+y+z)/(a+b+c)$

(vii) If U={1,2,3,4,5,6,7}, A={1,2,3,4} and B={3,4,5} then draw Venn diagrams for A' and B'

(viii) Resolve into partial fractions: $(3x-2)/((x+4)(3x^2+1))$

(ix) Domain of a binary relation $R=\{(x,y)|y+1=2x^2\}$ is set N. Find its range.

(x) Find the variance for the data 6,8,10,12,14

(xi) Find the length (l) of an arc of a circle of radius (r) 2cm and which subtend an angle ($\theta$) of $\pi/6$ radians.

(xii) Prove that $\tan\theta+\cot\theta=\sec\theta\operatorname{cosec}\theta$

## SECTION-C

**Maximum Marks:** 24

> Note: Attempt any three of the following questions. Each question carries 8 marks.

Q2. Prove that perpendicular from the centre of a circle on a chord bisects it.

Q3. Prove that if a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point.

Q4. Prove that if the angle subtended by two chords of a circle at the centre are equal in measures then the chords are equal in measures. Prove for only one circle.

Q5. Construct triangle ABC such that mAB=3cm, mAC=4cm, $m\angle A=60°$. Draw circumcircle to this triangle.